

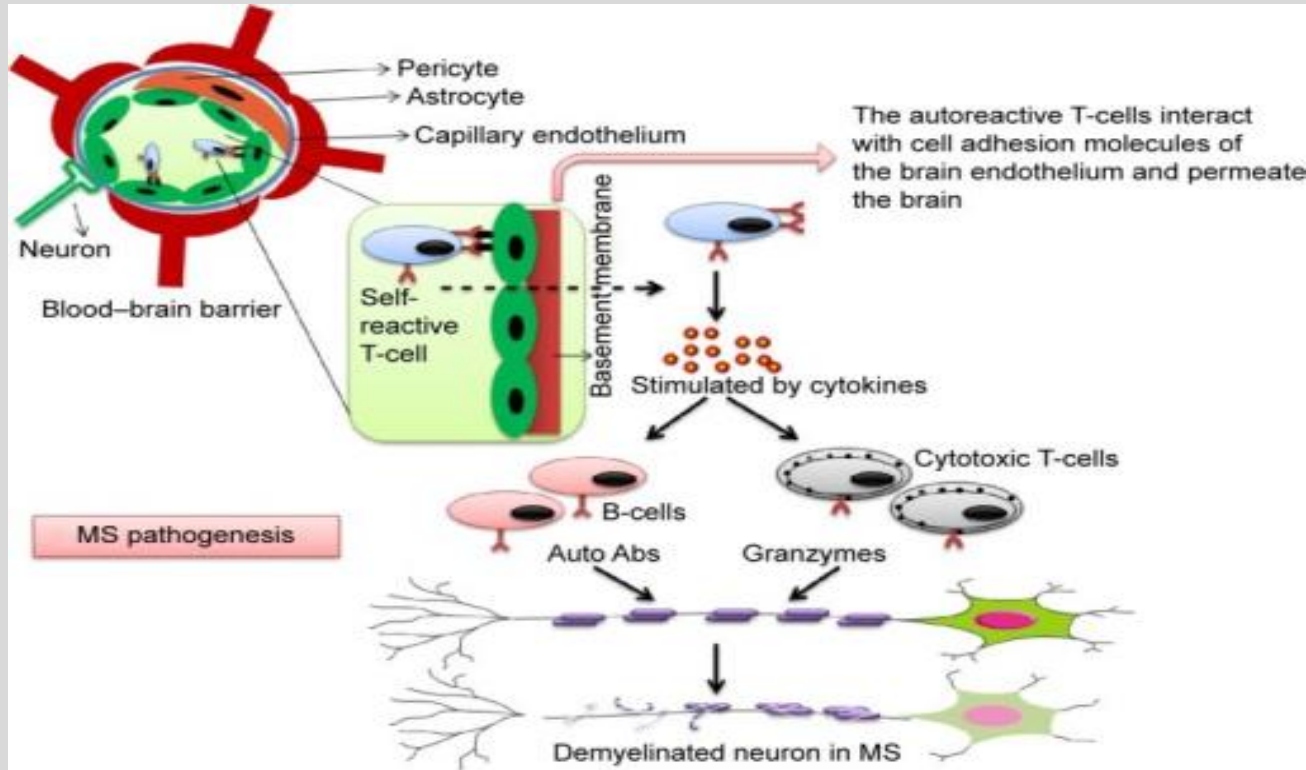
Session Objectives

- 1. Review the pathogenesis of Multiple Sclerosis (MS).**
- 2. Describe the drugs used for acute flare-ups of MS, mainly glucocorticoids. Identify adverse effects.**
- 3. Describe the use of immunotherapy for the treatment of MS. Focus on the pharmacological target and adverse effects.**
- 4. Describe supplemental treatments for MS. Include mechanisms and adverse effects.**

Pathogenesis of Multiple Sclerosis

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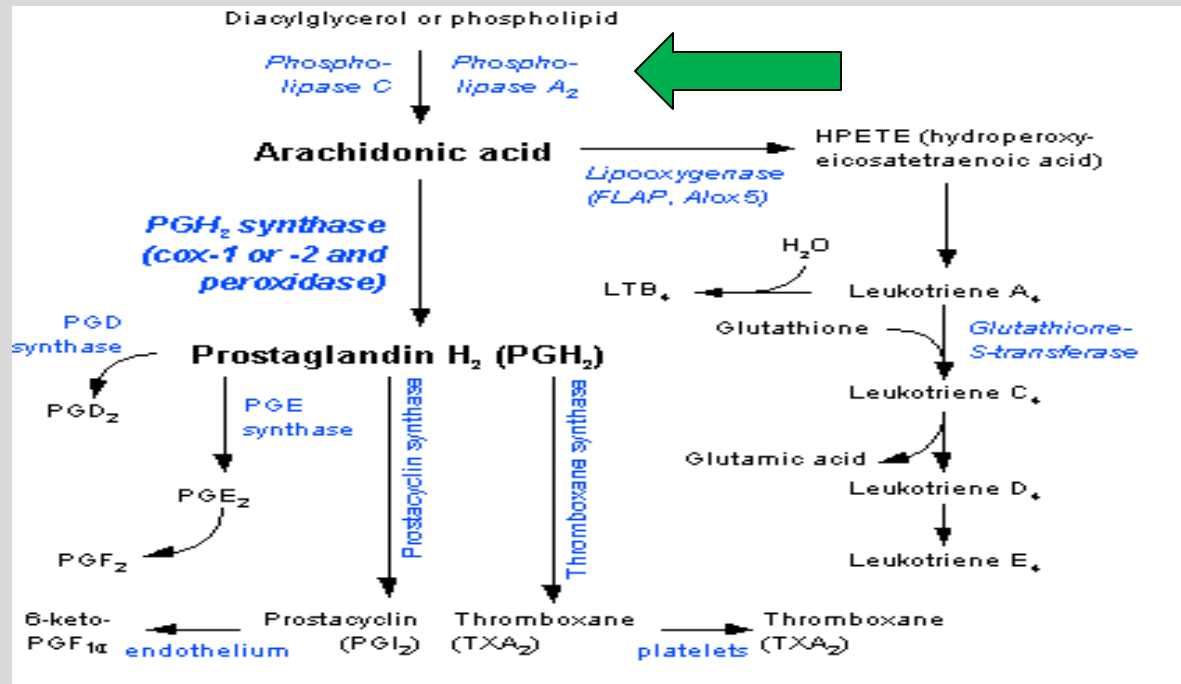
Glucocorticoids

- Stimulate synthesis of lipocortin

- Lipocortin inhibits Phospholipase A₂
 - Blocks starting point for prostaglandin, leukotriene synthesis

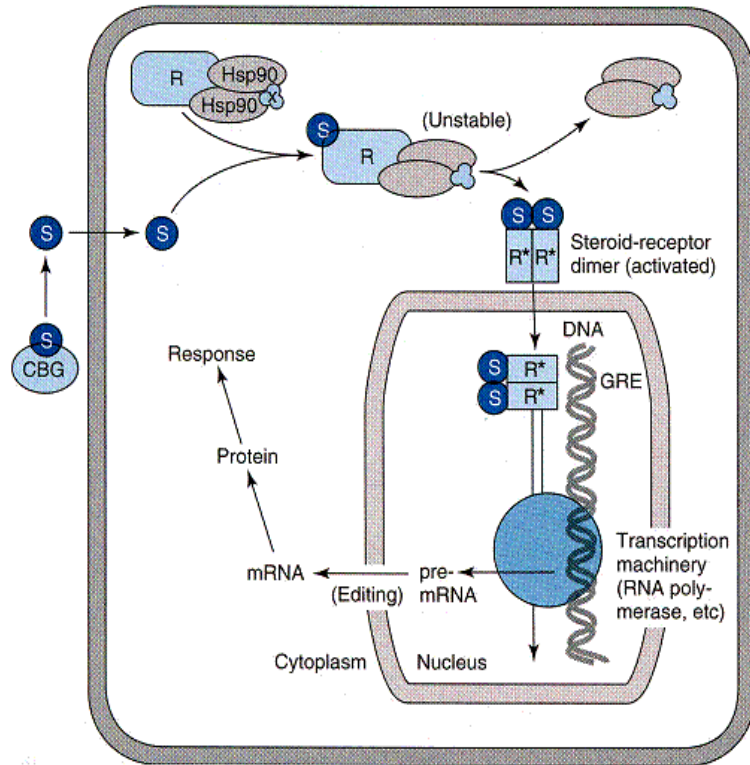
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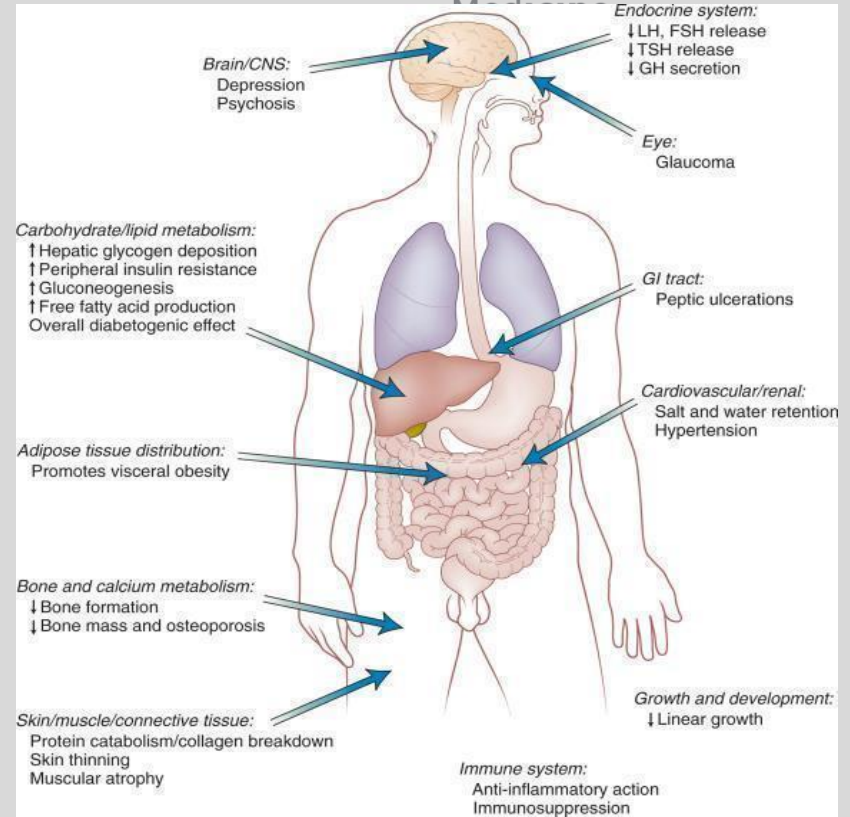
Glucocorticoids

HPA suppression: taper dose slowly



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Alemtuzumab: Anti-CD52

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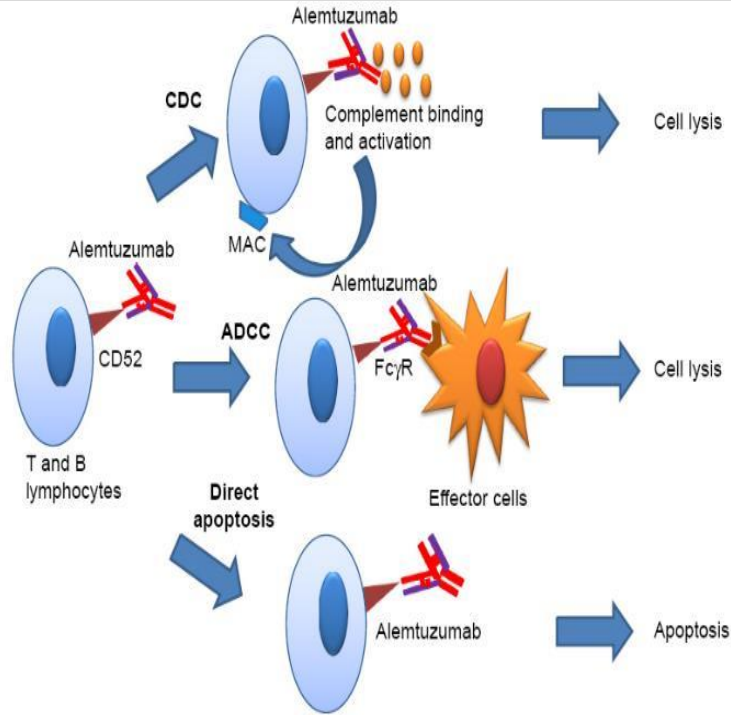


Figure 1 Alemtuzumab-mediated cytotoxicity and apoptosis of T- and B-lymphocytes.

Abbreviations: ADCC, antibody-dependent cell-mediated cytotoxicity; CDC, complement-dependent cytotoxicity; MAC, membrane attack complex; FcγR, Fc-gamma receptor.

AE:

- Pregnancy warning
- Depletes lymphocytes
- Autoimmune reactions
- Infusion reactions

Other uses:

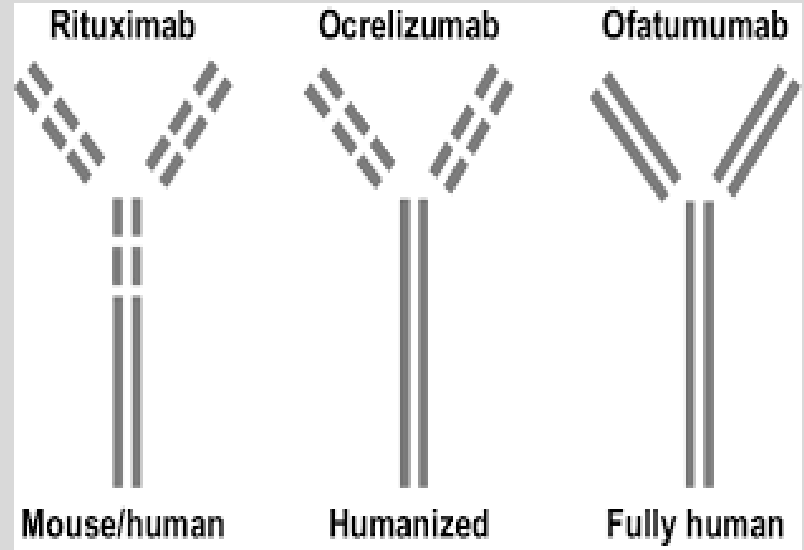
- Chronic lymphocytic leukemia (CLL)
- T-cell lymphoma.

Monoclonal Antibodies

- -umab = human (only human protein)
- -momab = mouse (murine)
- -ximab = chimeric (mix of mouse and human)
- -zumab = humanized (mix of mouse and human, less mouse protein compared to chimeric)

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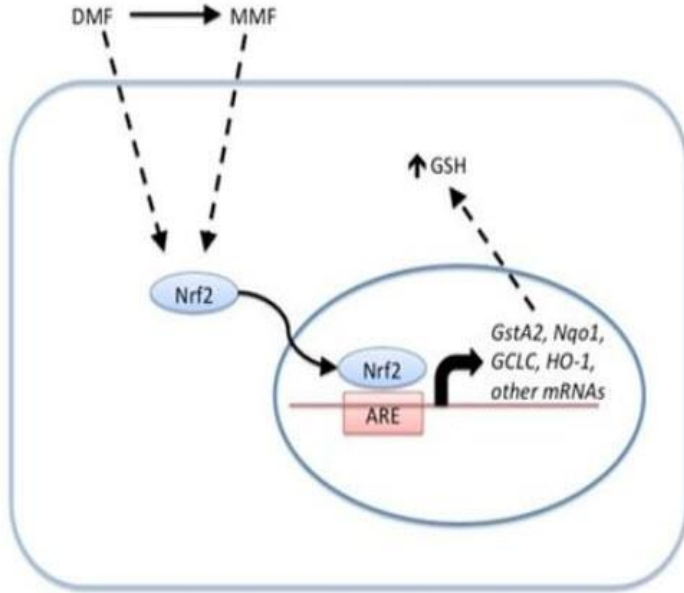
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Dimethyl fumarate

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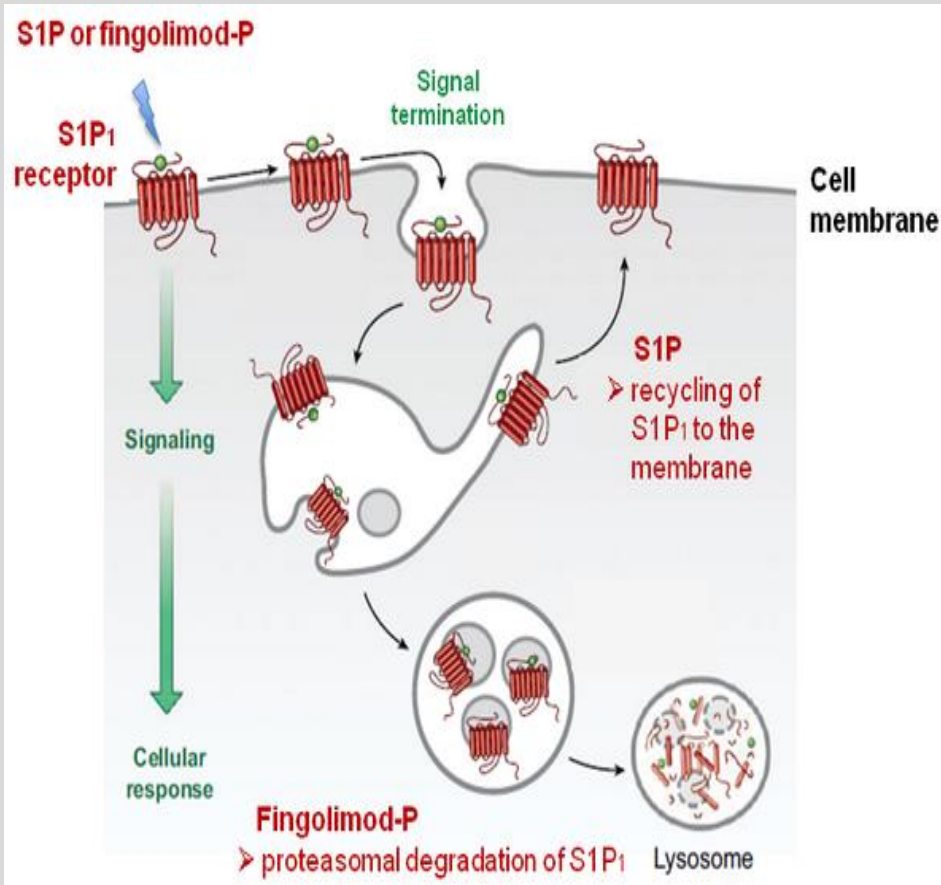


- **Nrf2 activator via metabolite monomethyl fumarate.**
- **AE: GI upset, flushing, Progressive multifocal leukoencephalopathy (PML), elevated liver enzymes, increased infection risk.**

Fingolimod

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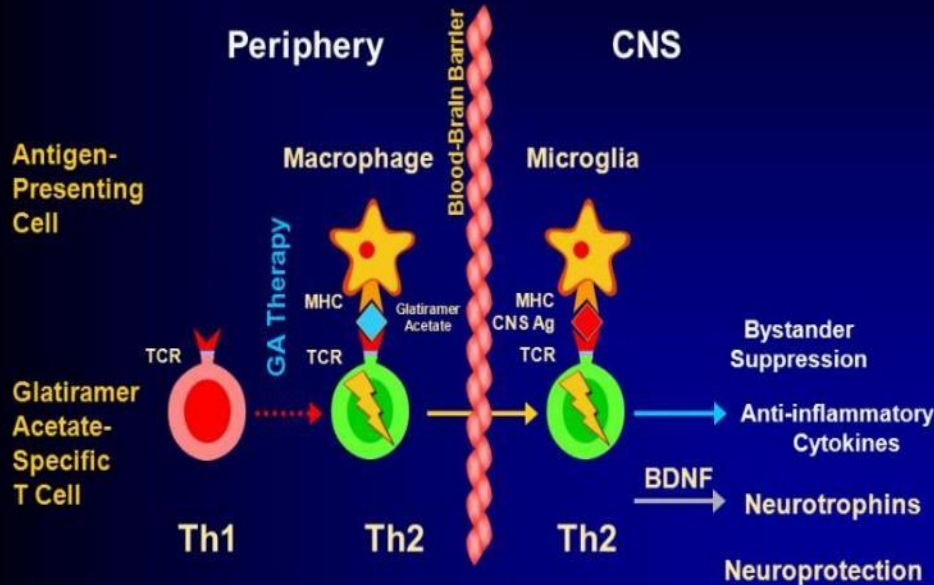
- Sequestering of lymphocytes
- AE: Headache, **bradycardia**, activation of infections, PML.
- Basal cell carcinoma
- Test VZV antibody
- Evaluate EKG (at first dose, re-initiation).

Glatiramer acetate

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Mode of Action of Glatiramer Acetate: Bystander Suppression and Neuroprotection



Adapted with permission from Neuhaus O et al. *Neurology*. 2001;56:702-708.
Ziemssen T et al. *Brain*. 2002;125:2381-2391.

- **AE: Injection site reactions**
- **Post-injection reaction: flush, SOB, palpitations ~occurs once.**
- **Lipoatrophy (SQ injection effect)**

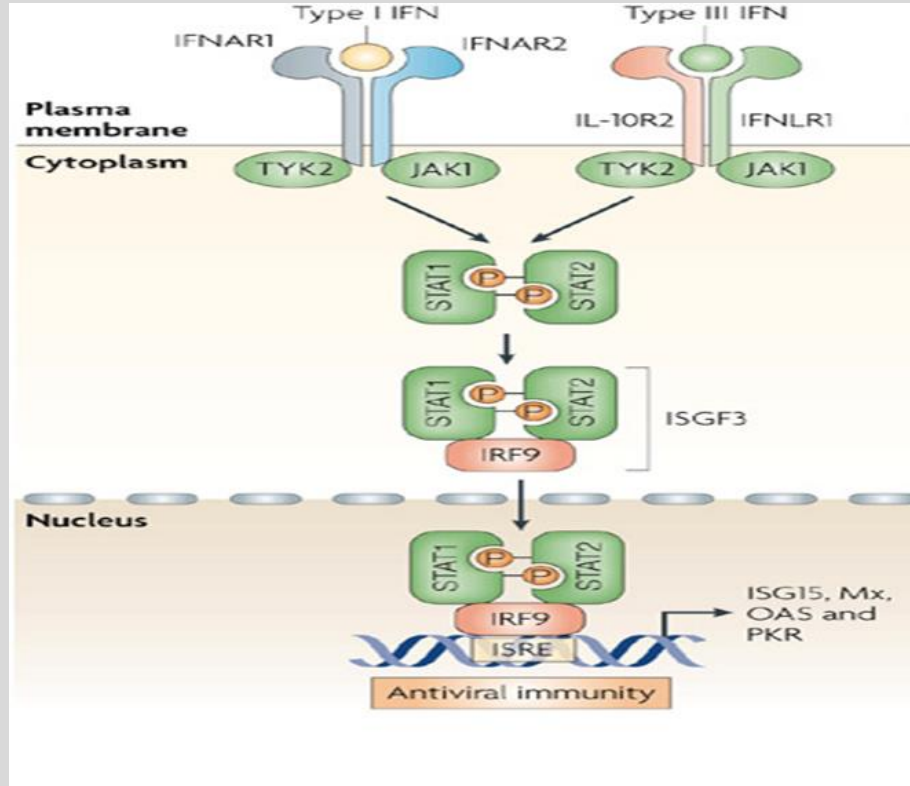
Why infusion reactions?

How to manage infusion reactions?

Interferon B1a

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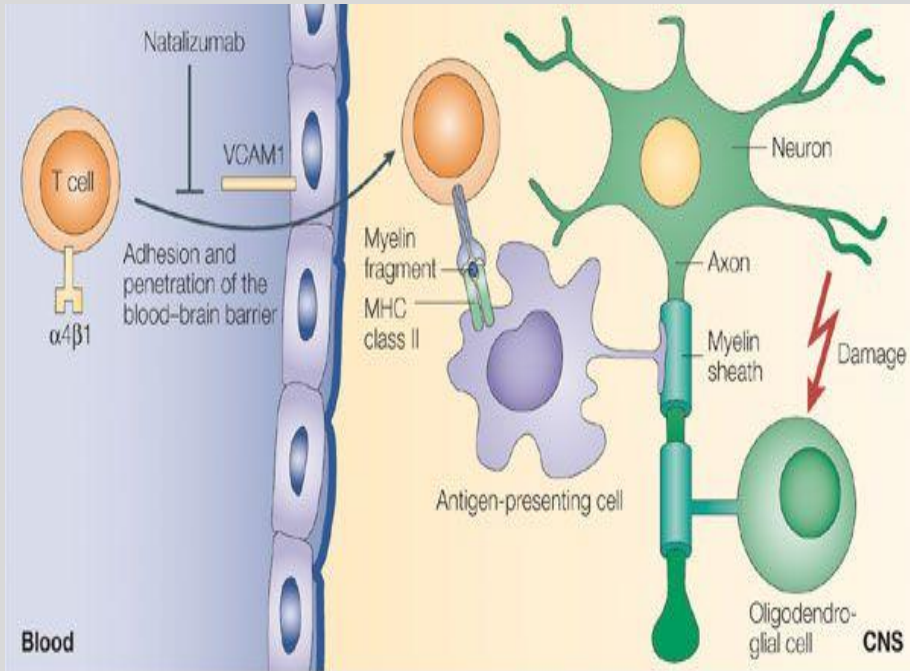


- Cytokine, Via JAK-STAT pathway
- MS: inhibits T-cell activation and migration across BBB.
- AE: flu-like symptoms, depression, injection reactions, anemia.
- Also treats hepatitis B, C.

Natalizumab

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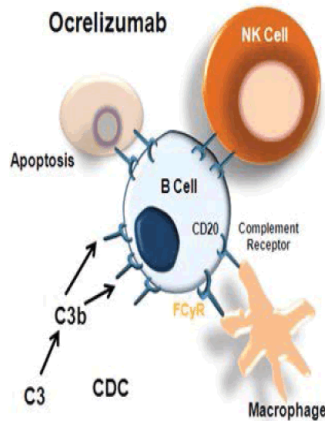
Nature Reviews | Drug Discovery

- **Binds ($\alpha 4$, $\beta 1$) receptor, stops T-lymphocyte adhesion to VCAM**
- **AE: infusion reactions, PML, hepatotoxicity, infection risk**
- **Also used for Crohn's disease**

Anti-CD20 drugs

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Antibody	Type	ADCC	CDC
Rituximab	Chimeric IgG1	++	++
Ocrelizumab	Fully humanized IgG1	+++	+/-

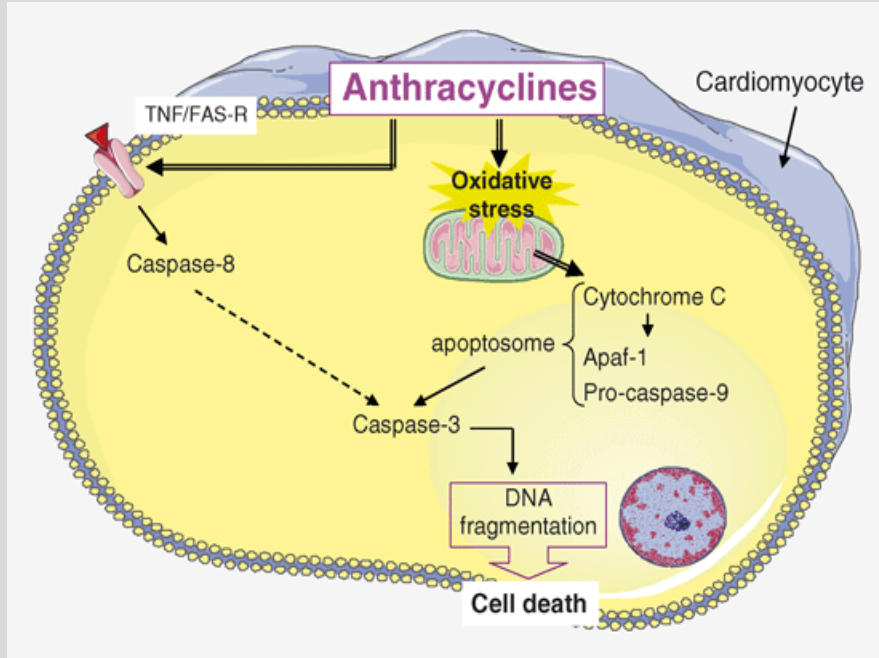
Figure 5: Ocrelizumab vs. Rituximab characteristics [17,18].

- **AE: infusion reactions, respiratory infections, activation of viral infections, such as hepatitis B**
- **Rituximab (B-cell lymphoma treatment)**
- **Ocrelizumab: Humanized**
- **Ofatumumab: Fully human**
- **Ublituximab: twice yearly infusion**

Mitoxantrone: Secondary Progressive MS

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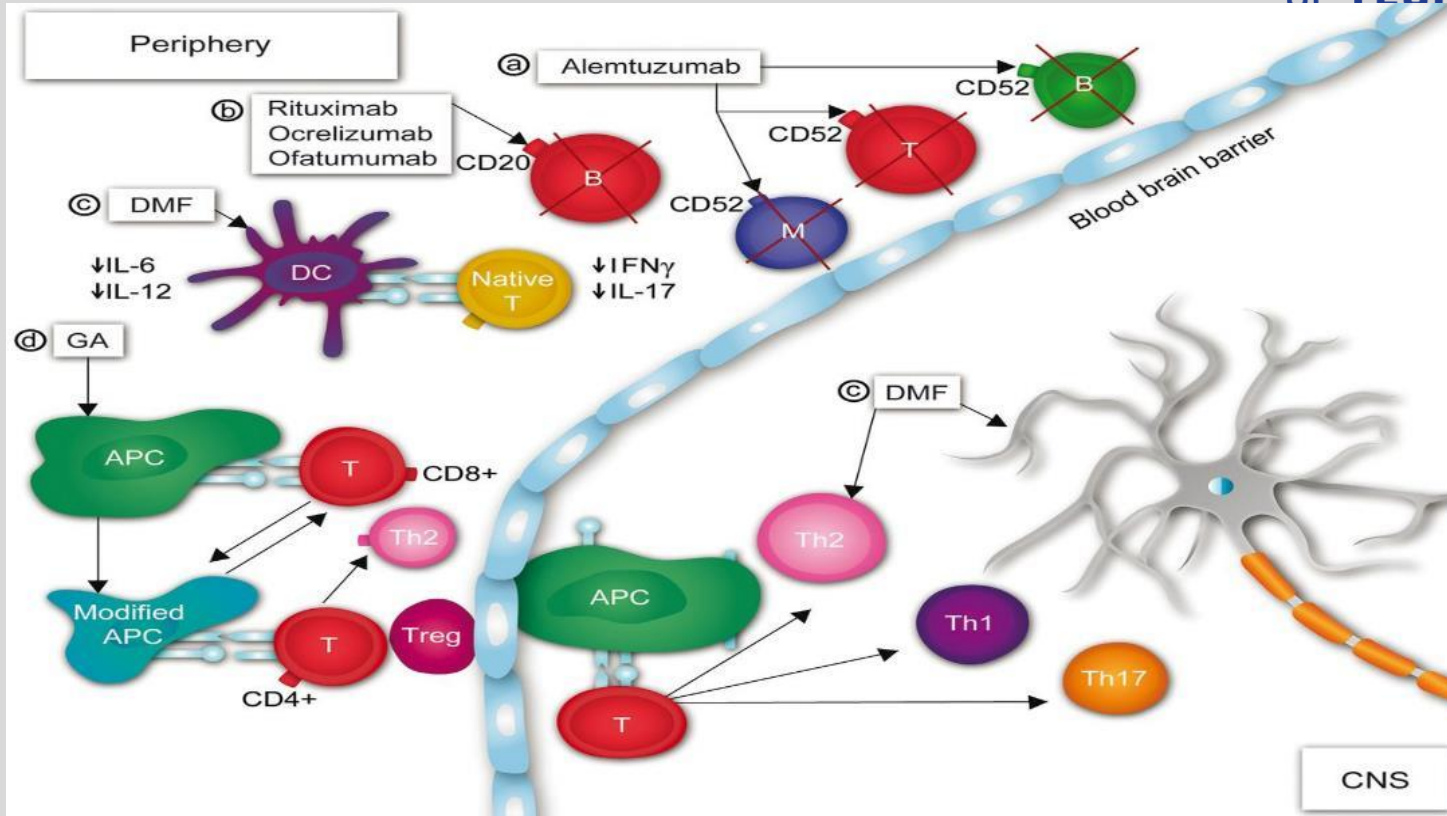


- Intercalating agent
- “Doxo**rub**icin” like. Anthracycline
- DNA strand scission via effects on topoisomerase II
- Binding to membranes alters fluidity
- Generation of oxygen radicals
- AE: cardio, N/V, alopecia, immunosuppression
- Also treats acute myeloid leukemia (AML), prostate, breast cancer

Summary: MS agents

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Osteopathic



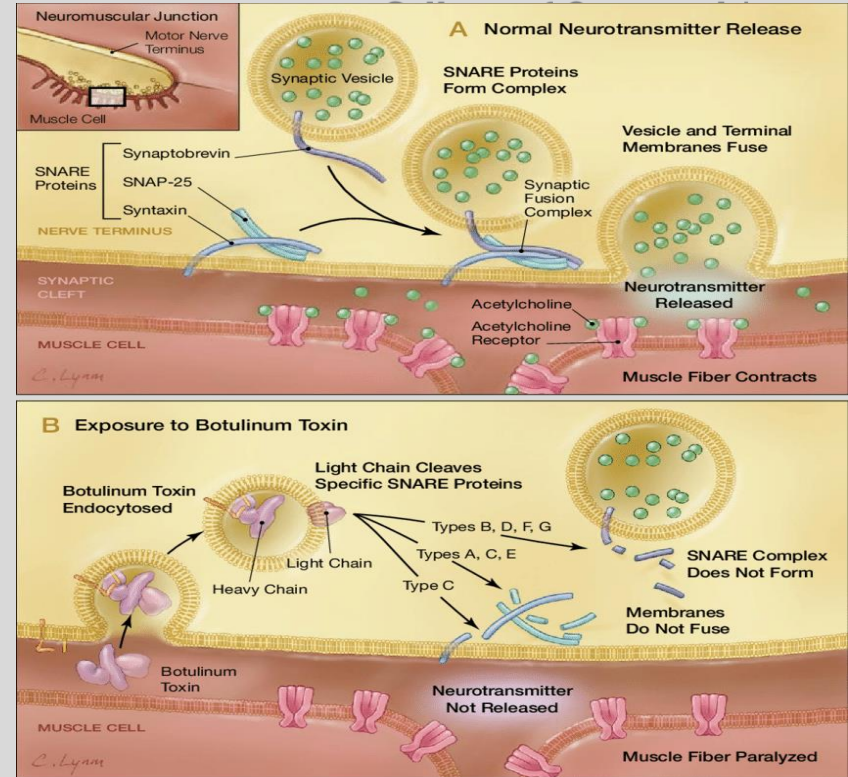
Incontinence

- M₃ blockers: oxybutynin, tolterodine, darifenacin
- Gq mechanism. AE??
- Botulinum toxin: cleaves SNARE protein (SNAP-25) responsible for vesicle fusion.

AE: Paralysis

Other uses: strabismus, blepharospasm, wrinkles, migraines, cervical dystonia

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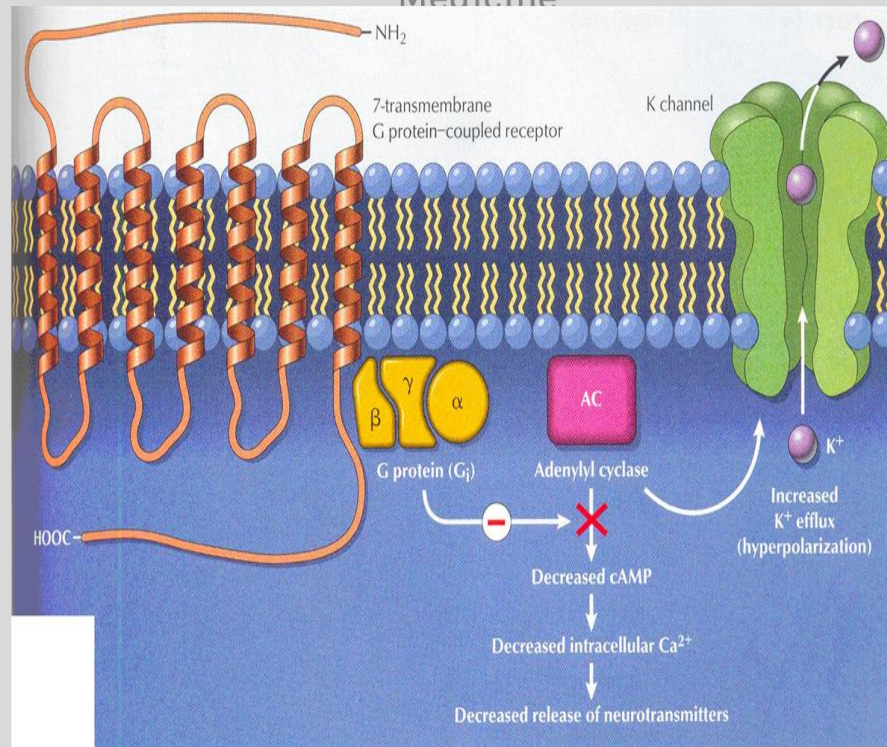
Spasticity

Goal: Reduce muscle tone

- Tizanidine (agonist of α_2 adrenergic receptor). Like clonidine
- Baclofen (GABA_B agonist): G protein (Gi) mechanism
- AE: CNS depression

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Summary Slide

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- Describe the drugs used for acute and relapsing MS. Focus on the pharmacological target and adverse effects.
- Describe supplemental treatments for MS. Include mechanisms and adverse effects

Core References:

1. Brunton and Knollman; Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 39: Immunotherapy for Multiple Sclerosis
2. Katzung and Vanderah; Basic & Clinical Pharmacology, 16e, 2024; Chapter 55: Immunopharmacology
3. Multiple Sclerosis drug summary and practice questions (Maria A. Pino, Ph.D)